Hena Singh

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Biomedical Engineering Junior at Texas A&M pursuing a Neuroscience minor with a strong interest in neurodegeneration and injury therapies. Research experience in the McCreedy Lab using mouse spinal cord injuries and Imaris image analysis. Seeking an internship in bioengineering and/or neuroscience for **Summer 2027**.

EDUCATION

Texas A&M University, College Station, Texas

BS in Biomedical Engineering, *GPA: 3.6/4.0* — August 2024 – May 2028

Minor in Neuroscience, *GPA: 4.0/4.0*

RELEVANT EXPERIENCE

McCreedy Lab (TAMU Spinal Cord Initiative), Texas A&M, College Station, TX

Undergraduate Research — May 2026 – Present

- Evaluated shrinkage of spinal cord during sciDISCO tissue clearing over a 5-day experiment; analyzed lengths of cords with ImageJ and discovered an average 17% diameter shrinkage
- Exported workflows for 97 SCI sections from Imaris 11.0 for 50+ hours
- Analyzed spinal cord vasculature on spinal cord-injured mice with Imaris 10.2 and 11.0
- Performed cryostat sectioning of spinal cords and applied over 80 delicate2µm slices to slides
- Examined samples perfused with WGA to determine vasculature repair after injury
- Received Mouse Handling Certification

TAMU Design Scholars, Texas A&M, College Station, TX

Time Blindness Project — January 2026 – May 2026

- Designed an LED bracelet prototype for patients with time blindness within a 3-month period
- 3D-printed a case with SolidWorks to hold a 16-LED NeoPixel ring that connected to an Adafruit microcontroller; USB connected to VS-Coded app that allowed user to adjust color of LEDs, durations of lights, and schedule of light pattern
- Researched neurological basis of time blindness and interviewed 10+ ADD/ADHD patients
- Applied Arduino, C++, and CAD/3D printing to code and design an LED bracelet

UT National Medical-Device-Make-A-Thon — March 6-8, 2026

- Designed a mucous extractor for neonatal patients in low-resource areas
- Applied SOLIDWORKS, biostatistics, respiratory physiology, & device design skills

Gaharwar Lab, Texas A&M, College Station, TX

Undergraduate Research — January 2025 – January 2026

- Shadowed 20+ experiments examining impact of MoS2 nanoparticles on macrophages and M0/M1/M2 monocytes involving PCR, western blotting, and spectrophotometer readings on Nanodrop
- Developed pipetting, recording data, and centrifuging skills

ADDITIONAL EXPERIENCE

Texas A&M University, College Station, TX

Engineering Events Student Assistant, Part-Time — September 2024 – October 2025

- Co-led 20+ departmental scholarship banquets, donor luncheons, and engineering socials
- Revised Student Handbook (80+ pages) for future student employees
- Utilized CVENT, Excel, Teamwork, Canva, Qualtrics

Public Volunteer at Aggieland Humane Society (2 hrs/week) — August 2026 – Present

BMES Marketing Committee Member — January 2026 – May 2026

SKILLS

- **Technical Skills:** Python, R, SolidWorks, LabVIEW, ImageJ, Imaris
- **Courses:** Physiology for Bioengineers, Circuits, Signals, and Systems, Imaging Living Systems